NOT MEASUREMENT SENSITIVE

MIL-G-762E
26 October 1990
SUPERSEDING
MIL-G-762D
6 April 1981

# MILITARY SPECIFICATION

## GRADERS, ROAD, PACKAGING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the minimum requirements for methods, materials, and procedures for the preservation, packing, and marking of road graders and their attachments.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION:

FEDERAL

| | |
|---|---|
| L-P-378 | - Plastic Sheet and Strip, Thin Gauge, Polyolefin. |
| TT-P-664 | - Primer, Coating, Alkyd, Corrosion-Inhibiting, Lead and Chromate Free, VOC-Compliant. |
| UU-T-81 | - Tags, Shipping and Stock. |
| PPP-B-601 | - Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | - Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | - Boxes, Shipping, Fiberboard. |
| PPP-B-1055 | - Barrier Material, Waterproofed, Flexible. |
| PPP-P-40 | - Packaging and Packing of Hand Tools. |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

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| MIL-P-116 | - Preservation, Methods of. |
| MIL-B-121 | - Barrier Material, Greaseproofed, Waterproofed, Flexible. |
| MIL-S-196 | - Support Items, Accessories and Kits, Mechanical, Packaging of. |
| MIL-L-2105 | - Lubricating Oil, Gear, Multi-Purpose. |
| MIL-C-5501 | - Cap and Plug, Protective, Dust and Moisture Seal. |
| MIL-E-10062 | - Engines: Preparation for Shipment and Storage of. |
| MIL-T-22085 | - Tape, Pressure-sensitive, Adhesive, Preservation and Sealing. |
| MIL-B-22191 | - Barrier Material, Transparent, Flexible, Heat Sealable. |
| MIL-L-23398 | - Lubricant, Solid Film, Air Cured, Corrosion Inhibiting. |
| MIL-L-46010 | - Lubricant, Solid Film, Heat-Cured, Corrosion-Inhibiting. |
| MIL-A-46153 | - Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package. |
| MIL-B-46176 | - Brake Fluid, Silicone, Automotive All Weather, Operational and Preservative, Metric. |

STANDARDS

MILITARY

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| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - Marking for Shipment and Storage. |

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

| | |
|---|---|
| D 3953 | - Strapping, Flat Steel and Seals. |
| D 4675 | - Selection and Use of Flat Strapping Materials. |

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)



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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be as specified herein and in applicable specifications and standards. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free of defects (see 6.5).

3.2 First article pack. Unless otherwise specified (see 6.2) the contractor shall furnish a first article pack for examination within the time frame specified (see 6.2) to prove, prior to starting production packaging, that the applied preservation, packing, and marking comply with the requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3). The first article pack may be accomplished utilizing either the first article grader or a production grader. If the first article model grader is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the first article grader and the production grader.

3.3 Preservation. Preservation shall be level A or C, as specified (see 6.2).

3.3.1 Level A.

3.3.1.1 Disassembly. Disassembly shall be the minimum necessary to safe-guard parts vulnerable to damage, pilferage, and loss. The extent of further disassembly shall be confined to the removal of protruding parts that would otherwise increase cubage; however, disassembly shall be limited to those parts that can be removed and reinstalled without special tools. Removed bolts, nuts, pins, screws, and washers shall be reinstalled in mating parts and secured to prevent their loss. Keys removed shall be secured to the keyway of the primary component or shaft. The extent of further disassembly shall be confined to that required by the transportability requirements of the end-item specification.

3.3.1.2 Matchmarking. Parts removed and mating parts on the graders and attachments shall be matchmarked when necessary to facilitate reassembly. Removed parts and mating parts shall be identified by cloth shipping tags conforming to UU-T-81, type A, style optional, and the marked tags attached to mating parts. The information on the cloth tags shall be applied with waterproof ink.

3.3.1.3 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

3.3.1.4 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically specified herein or in the end item specification shall be coated as follows.



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3.3.1.4.1 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

3.3.1.4.2 Machined surfaces. Exposed machined surfaces shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable with barrier material conforming to MIL-B-121, type I or II, grade A, class 2 or MIL-B-22191, type II. The material shall be secured in place with tape conforming to MIL-T-22085, type II.

3.3.1.5 Engines. Engines, engine components, and engine accessories shall be preserved in accordance with the level A requirements of MIL-E-10062, type II, method I.

3.3.1.5.1 Cooling systems. Engines with liquid cooling systems shall be preserved by the water and antifreeze procedure specified in MIL-E-10062, except the antifreeze shall conform to MIL-A-46153.

3.3.1.6 Instrument panels. Instruments, meters, and gages not protected by a metal cover or by the grader cab shall have the instrument glass or gage glass covered with barrier material conforming to MIL-B-121, type II, grade A, class 2 over which a fitted piece of hardboard or 1/4-inch-thick plywood shall be placed and secured with tape conforming to MIL-T-22085, type II.

3.3.1.7 Exposed gears. Exposed gears shall be coated with type P-1 preservative or with primer conforming to TT-P-664.

3.3.1.8 Enclosed gears. Gears operating on lubricating oil (SAE 10, 30, or 50) shall have the housing filled to the operating level with type P-10 preservative, type I, grade 10, 30, or 50, as applicable and tagged to indicate: "The preservative oil is adequate for operation. Do not drain until the first required lubrication change." Gears operating on gear lubricant (SAE 80 or 90) shall be filled to the operating level with lubricant conforming to MIL-L-2105, grade as applicable, and tagged to indicate: "The lubricant is good for operation. Do not drain until the first required lubrication change." Gears not operating on lubricating oil (SAE 10, 30, or 50) or lubricant (SAE 80 or 90) shall be filled to the operating level with the approved lubricant required for operation. The gear housings shall be tagged to indicate: "The housing is filled to the operating level with the applicable type lubricant. Do not drain prior to the first required lubrication change." The tags shall conform to UU-T-81, type A. The information on the tags shall be applied with waterproof ink.

3.3.1.9 Automatic transmissions and torque converters. The housings shall be filled to the operating level with the lubricant required for operation. The transmissions and torque converters shall be operated at sufficient engine speed to insure coating of all interior surfaces and parts with the lubricant. Openings into the housing shall be sealed with tape conforming to MIL-T-22085, type II. A tag shall be attached to the control lever indicating: "The housing is filled to the operating level with the approved lubricant and is good for operation. Do not operate engine until the tape has been removed from the openings." The tag shall conform to UU-T-81, type A. The information on the tag shall be applied with waterproof ink.



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3.3.1.10 Disc-type clutches. With the cover plates removed and the clutch engaged, all accessible metal interior components of the clutch shall be sprayed with a thin film of primer conforming to TT-P-664. When specified (see 6.2), the clutch shall be disassembled sufficiently to permit the application of a continuous thin film (max. 0.75 mil) of the same primer to friction surfaces of the pressure plates. Clutch control mechanisms not enclosed shall be coated with type P-1 preservative. After being sprayed, spring-loaded-type clutches shall have the clutch pedal secured in a partially disengaged position to eliminate contact between the disc facings and pressure plates. Snap-over-center and toggle-in-type clutches shall be completely disengaged. Cover plates shall be re-installed.

3.3.1.11 Exposed drive chains. Exposed drive chains shall be coated with enough type P-3 or P-9 preservative to insure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. After the excess preservative has drained, the entire chain and the unpainted surfaces of the sprocket shall be coated with type P-1 preservative.

3.3.1.12 Enclosed chains. Enclosed chains and chain housings shall be preserved and tagged as specified in 3.3.1.8 for enclosed gears.

3.3.1.13 Drive belts and pulleys. Drive belts shall be removed or released from tension. Removed belts shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3. Unpainted faces of pulley grooves shall be coated with primer conforming to TT-P-664. A tag conforming to UU-T-81, type A, shall be attached in a conspicuous location indicating: "Belts have been removed or released from tension. Install or tension as applicable, prior to operation". The information on the tag shall be applied with waterproof ink.

3.3.1.14 Drive shafts and universal joints. Unpainted surfaces of drive shafts and universal joints shall be coated with type P-1 preservative. Dust boots, if furnished, shall be reinstalled.

3.3.1.15 Flywheel and starter drives. When specified (see 6.2), flywheel ring gears and starter drive mechanisms shall be coated with type P-10 preservative, type I, grade 30.

3.3.1.16 Winterization system. Interior surfaces of the tanks, pumps, piping and burners requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-10 preservative, type I, grade 30. The threaded surfaces of fill caps and drain plugs shall be coated with the preservative and reinstalled.

3.3.1.17 Hydraulic systems (except hydraulic brakes or steering systems). The hydraulic fluid supply tanks shall be filled to the operating level with hydraulic fluid required for operation. The pistons shall be retracted as far as practicable into the cylinders and secured. When the pistons cannot be fully retracted, the exposed portions of the piston rods (ramshafts) shall be coated with type P-6 or P-11 preservative and the coated surfaces wrapped or covered with barrier-material conforming to MIL-B-121, type II, grade A, class 2, extending the wraps approximately 2 inches on the ram cylinders. The wraps shall be secured in place with tape conforming to MIL-T-22085, type II. When the pistons can be fully retracted, any remaining uncoated surfaces of the piston rods shall be coated with type P-1 preservative, with no wrapping required. The



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hydraulic control valves shall be secured in the neutral position and preserved as specified herein for piston rods. Hoses shall not be disconnected. Any unpainted exterior metal surfaces of the system, requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservative. A tag conforming to UU-T-81, type A, shall be attached to the control lever indicating: "The hydraulic supply tank is filled to the operating level with fluid required for operation. Do not drain." Waterproof ink shall be used for printing the information on the tag.

3.3.1.18 Brakes.

3.3.1.18.1 General. Exterior, unpainted, untreated, and threaded surfaces of cables, clevises, and linkages of service and parking brakes, requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservative. When specified (see 6.2), metal surfaces enclosed within brake drums (such as adjusting wedges, pins, eccentrics, cam levers, linkages, anchors, retracting springs) and the braking surfaces of the brake drums shall be sprayed with a continuous film of primer conforming to TT-P-664. The film shall be a minimum of 0.50 mil to a maximum of 0.75 mil thick as determined by an elcometer. Primer material containing lumps or skin, that will not return to a homogenous mixture upon stirring, shall not be used. The film shall be thoroughly dry prior to reinstalling the brake drums. A warning tag conforming to UU-T-81, type A, shall be securely attached in a conspicuous location in the operator's compartment, indicating "Brake drums and components preserved. Do not apply brakes when grader is being moved. Towing shall be accomplished by use of a rigid towbar or similar arrangement until this grader is issued to the user." Waterproof ink shall be used for the information marking on the tag. Note: Interior surfaces of brake drums shall not be preserved when the grader design incorporates a planetary drive with the wheel brake assembly and the surfaces are inaccessible by simply removing the brake drums.

3.3.1.18.2 Hydraulic brakes and steering systems.

3.3.1.18.2.1 Operating on nonpetroleum base hydraulic fluid. The complete hydraulic system of brakes and steering systems that operate on nonpetroleum base hydraulic fluids shall be filled to the operating level with preservative fluid conforming to MIL-B-46176 and the system actuated to insure coating of all interior surfaces with the preservative. A tag conforming to UU-T-81, type A, shall be attached to the controls indicating: "This system is filled with preservative fluid and is good for indefinite storage and for operation in ambient temperatures not lower than -10 °F. Prior to operation in ambient temperatures lower than -10 °F, the system shall be drained and refilled to the operating level with the hydraulic fluid required for operation." Waterproof ink shall be used for printing the information on the tag.

3.3.1.18.2.2 Operating on petroleum base hydraulic fluid. The complete hydraulic system of brakes and steering systems that operate on petroleum base fluid shall be filled to the operating level with the fluid required for operation and the systems actuated to insure coating of all interior surfaces with the fluid. A tag conforming to UU-T-81, type A, shall be attached to the system controls indicating: "This system is filled with the approved operating fluid." Waterproof ink shall be used for printing the information on the tag.



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3.3.1.18.3 Air-actuated brake systems. The air-actuated brake systems shall be drained. Interior surfaces of the air-supply tanks shall be fogged with type P-10 preservative, type I, grade 30. Excess preservative oil shall be drained. Threaded openings and the threads of drainplugs shall be coated with the preservative oil specified herein. The drainplugs shall be reinstalled. Air-line filters shall be drained and closed. The exhaust ports of relay emergency, quick-release, and relay valves not equipped with exhaust check valves shall be closed by inserting caps or plugs as applicable, conforming to MIL-C-5501 or by sealing the ports with tape conforming to MIL-T-22085, type II. Tags conforming to UU-T-81, type A, shall be attached to the valves and service lines indicating: "The plugs, caps, or tape must be removed from the exhaust ports and valves before this grader is placed in operation." Waterproof ink shall be used for printing the information on the tags.

3.3.1.19 Electrical components. Openings in sockets, coupling plugs, and switches shall be sealed with tape conforming to MIL-T-22085, type II.

3.3.1.20 Lamps, reflectors, and mirrors. Lamps, reflectors, and mirrors, if removed, shall be individually preserved in accordance with MIL-P-116, method III, using boxes conforming to PPP-B-636, W5c or W6c, style optional. Lamps, reflectors and mirrors not removed shall be covered with barrier material conforming to MIL-B-121, type II, grade A, class 2. The barrier material shall be secured in place with tape conforming to MIL-T-22085, type II.

3.3.1.21 Gear shifts and control levers. Unpainted metal surfaces of handles and levers shall be coated with type P-1 preservative. Composition and rubber handgrips, when not protected by a full cab, shall be removed if practical and preserved in accordance with MIL-P-116, method IC-1 or IC-3. When the handgrips are left in place they shall be wrapped with barrier material conforming to PPP-B-1055, class E-2, or black polyethylene conforming to L-P-378 (6 mil) and the wrap secured with tape conforming to MIL-T-22085, type II.

3.3.1.22 Seat backs, arm rests, and cushions. Seat backs, arm rests and cushions when not protected by a full cab, shall be removed if practical and individually wrapped with barrier material conforming to PPP-B-1055, class E-2 or black polyethylene conforming to L-P-378 (6 mil), and the wrap secured with tape conforming to MIL-T-22085, type II. When seat backs, arm rests, and cushions are left in place they shall be wrapped or covered with the barrier material specified herein and the wrap or cover secured in place with tape specified herein.

3.3.1.23 Windshield wiper arms and blades. Windshield wiper arms and blades shall be removed and preserved in accordance with MIL-P-116, method IC-1 or IC-3. The preserved wiper arms and blades shall be placed in the toolbox. The exposed wiper shaft shall be coated with type P-1 preservative.

3.3.1.24 Extra cutting edges and end bits. Unpainted surfaces of extra cutting edges and end bits shall be coated with type P-1 preservative. The extra cutting edges shall be secured to the grader (see 3.4.1.1).

3.3.1.25 Locks, keys, and key openings. Interiors of locks shall be coated with solid film lubricant conforming to MIL-L-46010 or MIL-L-23398. The locks shall be operated to insure penetration of the lubricant. Openings into locks (except padlocks) shall be sealed with tape conforming to MIL-T-22085, type II.



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Switch keys, cab door keys, and padlocks with keys shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3. The preserved keys and padlocks with keys for each grader shall be placed in the toolbox. A tag shall be attached in a conspicuous location in the operator's compartment indicating: "The keys for the grader are in the toolbox." For graders without cabs the information on the tags shall be applied with a waterproof ink.

3.3.1.26 Rubber tires. Tires mounted on the wheels (road tires) of the grader shall be inflated to maximum operating pressure for maximum load. Tires mounted on rims (spare tires) shall be inflated to two-thirds of the specified operating pressure.

3.3.1.27 Fire extinguishers. Unpainted exterior metal surfaces of fire extinguishers requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservative. Fire extinguishers enclosed with full cabs shall be secured in their mounting brackets. Fire extinguishers not enclosed within a full cab shall be individually placed in a box conforming to PPP-B-636, class weather-resistant. Blocking and cushioning shall be provided within the box to prevent movement or damage. Boxes shall be sealed in accordance with the appendix to the box specification, method V.

3.3.1.28 Maintenance tools. Maintenance tools shall be preserved in accordance with the level A preservation and packaging requirements of PPP-P-40.

3.3.1.29 Technical publications. Technical publications for each grader shall be preserved together in accordance with MIL-P-116, method IC-1 or IC-3.

3.3.1.30 Repair parts. Unless otherwise specified, the preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts. When specified (see 6.2), the repair parts shall be preserved in accordance with the level A requirements of MIL-S-196, or when parts are not specifically covered in MIL-S-196, an applicable sub-method of preservation of MIL-P-116 shall be used.

3.3.1.31 Other components. Other components not specifically mentioned herein requiring protection from corrosion or physical or mechanical damage shall be preserved as specified for components of similar design and construction.

3.3.1.32 Consolidation.

3.3.1.32.1 Components. Components for each grader, except extra cutting edges, shall be consolidated together in boxes conforming to PPP-B-621, class 2, style optional; or PPP-B-601, overseas type, style optional. The boxes shall be close-fitting. The boxes shall be strapped in accordance with the appendix to the applicable box specification.

3.3.1.32.2 Repair parts. Repair parts for each grader or for attachments not attached to the grader shall be consolidated together in boxes as specified in 3.3.1.32.1.

3.3.1.32.3 Maintenance tools and technical publications. Maintenance tools and technical publications for each grader shall be placed in the toolbox.



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3.3.1.33 Toolbox. The toolbox lid shall be closed and secured to prevent pilferage.

3.3.1.34 Attachments. Attachments not attached to the grader, such as snowplows, rollers, scarifiers, backhoes, bulldozer blade, loaders, and scrapers, shall have the hydraulic systems, gears, chains, and other components preserved as specified herein for similar components of attachments installed on the grader. Hydraulic lines and other openings into the attachments shall be sealed by dummy couplers if so equipped, or with caps or plugs conforming to MIL-C-5501, or with tape conforming to MIL-T-22085, type II. Consolidation of disassembled components for the attachment shall be in boxes specified in 3.3.1.32.1. The consolidated containers shall be secured to the attachment for which intended with flat steel strapping conforming to ASTM D 3953, zinc-coated, type 1 or 2, size as applicable, in a manner not to interfere with lifting or moving the attachment and ASTM D 4675.

3.3.2 Level C. Each complete grader, attachments, repair parts, maintenance tools, and technical publications shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage. The grader shall be lubricated for operation service as required in the operator's manual.

3.4 Packing. Packing shall be level A or C as specified (see 6.2).

3.4.1 Level A. The graders preserved as specified in 3.3, shall be prepared for mobile (unboxed) shipment as follows.

3.4.1.1 Components and consolidated containers. Preserved and unpreserved components, consolidated containers, and extra cutting edges shall be secured on the grader with strapping conforming to ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675. The items shall be secured in such a manner that the minimum cube will be realized, wheels shall be free to rotate, and arrangement shall not interfere with towing or lifting the grader with slings.

3.4.1.2 Moldboard and circle. The circle shall be centered under the main grader frame and the moldboard shall be turned so that the ends do not extend beyond the outside of the tires and wheels. The circle and moldboard shall be raised to the maximum height and locked in position, and for hydraulically controlled graders shall be additionally secured to prevent movement by tying to the grader frame with wire rope, size as applicable, secured with wire rope clips or with strapping conforming to ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675.

3.4.1.3 Scarifier. The scarifier shall be raised to its maximum height and locked, and for hydraulically controlled graders shall be additionally secured by tying to the grader frame with wire rope, size as applicable, secured with wire rope clips or with strapping conforming to ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675.

3.4.1.4 Leaning-wheel graders. Leaning wheels shall be positioned vertically and locked in place.

3.4.1.5 Engine housings. The housing doors shall be closed and secured. Louvers and other small openings shall be sealed with tape conforming to



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MIL-T-22085, type II. Engines not completely enclosed within housings shall be covered with barrier material conforming to PPP-B-1055, class E-2 secured with tape as specified herein. When specified (see 6.2), radiator fronts shall be covered with barrier material specified herein and secured with tape as specified herein.

3.4.1.6 Cabs. Cab windows and doors shall be closed. When specified (see 6.2), windshields and glass windows shall be protected from breakage by covering each glass with a fitted piece of hardboard, or with 1/4-inch-thick plywood or a sheet metal panel secured in place with bolts or with steel strapping conforming to ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675. There shall be not less than 1 inch clearance between the panels and the glass. The panels shall be painted the same color as the cab. Small openings into the cab shall be sealed with tape conforming to MIL-T-22085, type II.

3.4.1.7 Attachments. Attachments not attached to the grader shall be shipped unboxed. Controls shall be locked or secured in the neutral position. Loose ends of hoses shall be secured to the attachment in a manner to prevent damage. The ends of hoses shall be sealed with plastic or metal caps or plugs as applicable.

3.4.2 Level C. The graders, components, attachments, repair parts, tools, and publications preserved as specified in 3.3 shall be prepared for mobile (unboxed) shipment in a manner which will ensure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

3.5 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

3.6 Depreservation guides. Two depreservation guides shall be prepared and placed in waterproof envelopes marked "Depreservation Guide." One of the guides shall be secured to the steering wheel or steering controls of the grader. The other copy shall be placed in the grader toolbox. Unless otherwise specified (see 6.2), "DA Form 2258 - Depreservation Guide for Vehicles and Equipment" shall be used.

3.7 Workmanship. Workmanship shall be such as to provide maximum protection to the graders by application of the requirements specified herein, to prevent corrosion, deterioration, and mechanical or physical damage during handling, shipment, and storage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Material inspection. The contractor is responsible for insuring that supplies and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article pack inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article pack inspection.

4.3.1 Examination. The first article pack shall be examined for the defects specified in 4.4.4. Examination shall be in three stages as specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Inspection stages.

a. The first stage shall include inspection of procedures, containers, materials, methods, and marking prior to enclosing the loose components in consolidated containers.
b. The second stage shall include inspection of the marking and closure of the consolidated containers.
c. The third stage shall include inspection of the procedures methods, materials, and marking of the complete mobile (unboxed) grader as prepared for shipment. This stage shall include arrangement of consolidated containers and components required to be secured to the grader for shipment.

4.4.2 Unit of product. For the purpose of inspection a complete unit prepared for any applicable inspection stage shall be considered a unit of product.

4.4.3 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Samples shall be selected using MIL-STD-105, tables I and IIa.

4.4.4 Examination. Samples selected in accordance with 4.4.3 shall be examined for the defects marked "X" for the applicable level in table I. Presence of one or more defects shall be cause for rejection.



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TABLE I. Defects.

| MAJOR | DEFECTS | LEVEL A | LEVEL C |
|---|---|---|---|
| 101. | Materials, methods, and containers, not as specified. Each incorrect material, each incorrect method, and each incorrect container shall constitute one defect, (3.1, 3.3, 3.4). | X | - |
| 102. | Disassembly not as specified, (3.3.1.1). | X | - |
| 103. | Matchmarking not as specified, (3.3.1.2) | X | - |
| 104. | Information on tags not applied with waterproof ink, (3.3.1.2, 3.3.1.8, 3.3.1.9, 3.3.1.13, 3.3.1.17, 3.3.1.18.1, 3.3.1.18.2.1, 3.3.1.18.2.2, 3.3.1.18.3, 3.3.1.25). | X | - |
| 105. | Unprotected surfaces not coated with preservative as specified (3.3.1.4) | X | - |
| 106. | Coated surfaces not wrapped or covered as specified, (3.3.1.4) | X | - |
| 107. | Engines and engine components not preserved as specified (3.3.1.5, 3.3.2.1). | X | - |
| 108. | Instrument panels not protected as specified (3.3.1.6) | X | - |
| 109. | Exposed gears not coated with preservative or with primer as specified (3.3.1.7). | X | - |
| 110. | Enclosed gear housing not filled with preservative oils or with operational lubricant as specified and the housings not tagged as specified (3.3.1.8) | X | - |
| 111. | Automatic transmissions and torque converters not filled with operational lubricant and the control levers not tagged as specified (3.3.1.9) | X | - |
| 112. | Accessible interior metal surfaces of clutches not coated with primer as specified (3.3.1.10) | X | - |
| 113. | Clutches not completely disengaged or completely disengaged as applicable (3.3.1.10) | X | - |
| 114. | Exposed drive chains not coated with preservative as specified (3.3.1.11). | X | - |
| 115. | Enclosed chain housings not filled with preservative oil or not filled with operational lubricant as applicable and the housings not tagged as specified (3.3.1.12). | X | - |
| 116. | Drive belts not removed or released from tension as specified and not tagged as specified (3.3.1.23) | X | - |
| 117. | Unpainted surfaces of pulley grooves not coated with primer as specified (3.3.1.13). | X | - |
| 118. | Drive shafts and universal joints not coated with preservative as specified and dust boots not reinstalled as specified (3.3.1.14). | X | - |
| 119. | Winterization system components not preserved as specified (3.3.1.16). | X | - |
| 120. | Hydraulic systems not filled with fluid as specified (3.3.1.17). | X | - |



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TABLE I. Defects. (continued)

| MAJOR | DEFECTS | LEVEL A | LEVEL C |
|---|---|---|---|
| 121. | Exposed portions of hydraulic piston rods not preserved as specified and the preserved surfaces not wrapped when required (3.3.1.17). | X | - |
| 122. | Hydraulic systems not tagged as specified (3.3.1.17) | X | - |
| 123. | Surfaces of brake components enclosed within brake drums and the braking surfaces of brake drums not coated with primer when specified (3.3.1.18.1). | X | - |
| 124. | Exterior surfaces of brakes not coated with preservative as specified (3.3.1.18.1). | X | - |
| 125. | Tag not attached to indicate brakes have been preserved (3.3.1.18.1, 3.3.1.18.2.1, 3.3.1.18.2.2). | X | - |
| 126. | Hydraulic brake systems operating on nonpetroleum base fluid not filled with preservative fluid as specified (3.3.1.18.2.1). | X | - |
| 127. | Hydraulic brake systems operating on petroleum base fluid not filled to the operating level with the fluid required for operation (3.3.1.18.2.2). | X | - |
| 128. | Air-actuated brake system supply tank not fogged with preservative oil as specified, excess preservative not drained, ports and valves not sealed and tagged as specified (3.3.1.18.3). | X | - |
| 129. | Openings into electrical sockets, coupling plugs, and switches not sealed as specified (3.3.1.19). | X | - |
| 130. | Lamps, reflectors, and rear vision mirrors not preserved or not covered as applicable (3.3.1.20). | X | - |
| 131. | Composition or rubber handgrips not preserved or not wrapped as applicable (3.3.1.21). | X | - |
| 132. | Seat backs, armrests, and cushions not wrapped or not covered as specified (3.3.1.22). | X | - |
| 133. | Windshield wiper arms and blades not removed, preserved and placed in tool box as specified (3.3.1.23). | X | - |
| 134. | Interior of locks not coated with lubricant and openings into locks not sealed as specified (3.3.1.25). | X | - |
| 135. | Key and padlocks not placed in toolbox as specified (3.3.1.25). | X | - |
| 136. | Tag not attached within operators compartment to indicate location of keys (3.3.1.25). | X | - |
| 137. | Rubber tires not inflated as specified (3.3.1.26) | X | - |
| 138. | Unpainted exterior surfaces of fire extinguishers not coated with preservative, if required (3.3.1.27). | X | - |
| 139. | Fire extinguishers not placed in box as specified, if required or secured in mounting brackets (3.3.1.27) | X | - |
| 140. | Maintenance tools not preserved in accordance with the referenced document (3.3.1.28). | X | - |
| 141. | Technical publications not preserved as specified (3.3.1.29). | X | - |
| 142. | Repair parts not preserved in accordance with the referenced document (3.3.1.30). | X | - |



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TABLE I. Defects. (continued)

| MAJOR | DEFECTS | LEVEL A | LEVEL C |
|---|---|---|---|
| 143. | Consolidation not as specified (3.3.1.32) | X | - |
| 144. | Toolbox lids not secured to prevent pilferage (3.3.1.33). | X | - |
| 145. | Consolidated containers not secured on the grader in the manner specified (3.4.1.1). | X | - |
| 146. | Strapping not as specified (3.4.1.1, 3.4.1.6). | X | - |
| 147 | Moldboard and circle not positioned and secured as specified (3.4.1.2). | X | - |
| 148. | Scarifiers not positioned, not locked, and not secured as specified (3.4.1.3). | X | - |
| 149. | Leaning wheels not positioned as specified (3.4.1.4). | X | - |
| 150. | Engines not protected by a shroud when not fully enclosed (3.4.1.5). | X | - |
| 151. | Radiator fronts not covered as specified, when required (3.4.1.5). | X | - |
| 152. | Windshields and window glass not protected as specified, when required (3.4.1.6). | X | - |
| 153. | Marking missing, illegible, incomplete, incorrect (3.5). | X | X |
| 154. | Depreservation guides not prepared and stowed as specified (3.6). | X | X |
| 155. | Workmanship not as specified (see 3.7). | X | X |

5. PACKAGING

(This section not applicable to this specification.)

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. It is intended that the requirements of this specification be used for reference in section 5 of equipment specifications, for direct reference in contract or orders, and can be used as a guide in the preparation of packaging data sheets covering preservation, packing, and marking of road graders.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article pack is not required (see 3.2).
d. Time frame required for submission of first article pack (see 3.2).
e. Level of preservation and packing required (see 3.3 and 3.4).



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f. When friction surfaces of clutch pressure plates are to be preserved (see 3.3.1.10).
g. When flywheel ring gears and starter drive mechanism are to be preserved (3.3.1.15).
h. When brake components enclosed within brake drums and the braking surfaces of brake drums are to be preserved (see 3.3.1.18.1).
i. Applicable document to be used for preserving repair parts (see 3.3.1.30)
j. When radiator fronts are to be protected (see 3.4.1.5).
k. When windshields and window glass is to be protected (see 3.4.1.6).
l. When other than DA Form 2258 is to be used (see 3.6).
m. Any special marking (see 3.5).

6.3 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the graders and components in accordance with this specification.

6.4 Depreservation guides. The contracting officer should arrange to furnish the necessary copies of DA Form 2258, when requested by the contractor (see 3.6).

6.5 Recycled materials. It is encouraged that recycled materials be used, when practical, as long as it meets the requirements of this specification (see 3.1).

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activity:
Army - AT, SM
Air Force - 84

User activity:
Army - CE

Preparing activity:
Army - ME

Project 3805-0138



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-G-762E | 901026 |

3. DOCUMENT TITLE
Graders, Road, Packaging Of

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | c. TELEPHONE (Include Area Code) (1) Commercial / (2) AUTOVON | d. DATE SUBMITTED (YYMMDD) |
|---|---|---|---|
| | | | |

7. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (Include Area Code) (1) Commercial | (2) AUTOVON |
|---|---|---|
| | (703) 664-5717 | 354-5717 |

| c. ADDRESS (Include Zip Code) | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: |
|---|---|
| US Army Belvoir RDE Center<br>ATTN: STRBE-TSE<br>Ft. Belvoir, VA 22060-5606 | Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |

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